BERZIET UNIVERSITY
FACULTY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER ENGINEERING

ENCS5341

MACHINE LEARNING AND DATA SCIENCE

Assignment#1:

Data Preprocessing & Exploratory Data Analysis (EDA)

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Abstract

The assignment is based on the initial phases of the data science step - data preprocessing and exploratory data analysis (EDA), in a synthetic dataset of customers. The data is a simulation of real-world customer data, such as demographic, behavioral and subscription-related data and the aim of the data analysis is to know what factors contribute to customer churn. The dataset has undergone preprocessing steps such as handling missing values, detecting and treating outliers, and applying feature scaling. Besides this, statistical analysis and visual exploration is done in order to discover patterns, correlations, and possible predictors of churn. In this assignment, Python libraries, including pandas, matplotlib, and seaborn, are used to handle data and visualize it properly. The results give useful clues on the customer retention behavior and give a clean and well-organized data that can be used later in machine learning.

Procedure & Analysis: Steps for the Assignment:

1. Data Loading and Initial Inspection:

The data was loaded into a pandas dataframe which was subjected to analysis. The first records were made with the help of. head() to verify the existence of such key features as Age, Income, Tenure, SupportCalls, and ChurnStatus. The summary in the form of the .info() revealed that there were 3,500 rows in 8 columns; the missing values were in Age, Income, Tenure and SupportCalls. The results of the .describe() command suggested that the average age of the participants was 43.6 years, the average income was approximately 140,686 and the churn rate was really low, 4.5. All in all, the analysis indicated a rather thorough dataset with some values missing and possible outliers that need additional preprocessing.

2. Handling Missing Data:

Missing values were first identified using the .isnull().sum() function which revealed gaps in several numerical columns (Age, Income, Tenure, and SupportCalls) with approximately 170-175 missing values per column.

Approaches used: Imputation - Median and Mode

Justification:

In order to deal with these missing values, the numerical features were imputed using median, as it is more resistant to outliers and better preserves the central tendency of the data compared to mean imputation which would skew the mean if used. With categorical value, entries are replaced with the mode (the most frequent value) in order to have some consistency and avoids biasing the dataset with rare categories. This approach ensures that the imputed values reflect the typical patterns in the dataset rather than extreme or unusual entries. After imputation, all missing values were successfully filled, resulting in a clean dataset with no remaining null entries, ensuring data completeness and reliability for subsequent analysis.

Note: *Figure1 and Figure2* shown below illustrate sample data, while tables in the Jupyter Notebook display a subset before and after missing value imputation, with imputed values highlighted in yellow for clarity.

Check Missing Values:

```
CustomerID        0
Age             175
Gender            0
Income          172
Tenure          175
ProductType       0
SupportCalls    171
ChurnStatus       0
dtype: int64
```

Missing Data before Handling (highlighted in yellow):

[9]:

	CustomerID	Age	Gender	Income	Tenure	ProductType	SupportCalls	ChurnStatus
12	CUST0012	NaN	1	111551.0	NaN	0	3.0	0
13	CUST0013	53.0	1	34583.0	8.0	0	1.0	0
14	CUST0014	57.0	1	NaN	7.0	0	150.0	0

Figure 1: Missing Data before Handling

Check Missing Values:

```
CustomerID      0
Age             0
Gender          0
Income          0
Tenure          0
ProductType     0
SupportCalls    0
ChurnStatus     0
dtype: int64
```

Handled Missing Values (highlighted in yellow):

[34]:

	CustomerID	Age	Gender	Income	Tenure	ProductType	SupportCalls	ChurnStatus
12	CUST0012	43.0	1	111551.0	5.0	0	3.0	0
13	CUST0013	53.0	1	34583.0	8.0	0	1.0	0
14	CUST0014	57.0	1	89532.5	7.0	0	150.0	0

Figure 2: Handled Missing Data using Median for numeric values and Mode for categorical values

3. Handling Outliers

The Interquartile Range (IQR) strategy with Median replacement was used to detect outliers:

❖ Justification for Outlier Handling Approach

The Interquartile Range (IQR) strategy was used to detect outliers because it is a robust statistical method that focuses on the middle 50% of the data and is not influenced by extreme values. Once outliers were identified, they were replaced with the median rather than removed or replaced with the mean. The median is resistant to the influence of extreme values, helping maintain the central tendency and overall shape of the data distribution. This approach ensures that valuable data points are not lost while preventing extreme values from skewing model performance or biasing statistical results. As a result, the dataset remains consistent, reliable, and better suited for accurate machine learning analysis.

3.1. Identifying Outliers

Outliers were first identified using box plots, which visually highlight unusually high or low values in numerical features. For each selected feature, a box plot was generated as shown in *Figure3* below to inspect the data spread and detect any points falling outside the typical range. These extreme points represent potential outliers that may negatively affect model performance and statistical analysis.

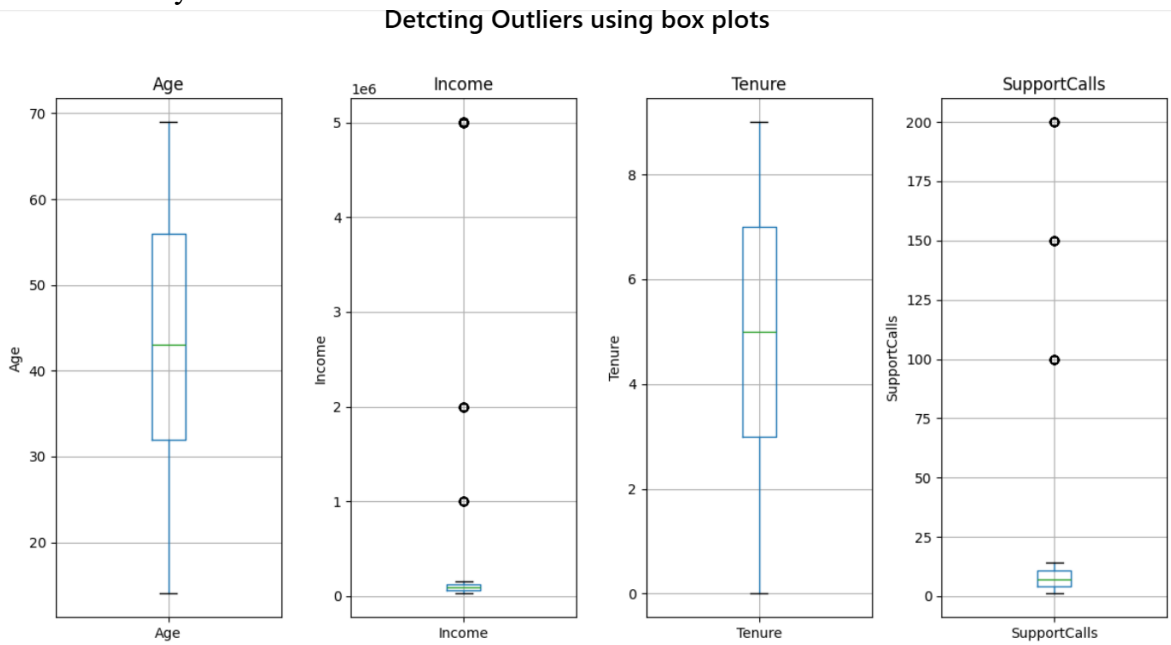


Figure 3: Detecting Outliers using box plots

3.2. Handling Outliers using IQR strategy with Median Replacement

To avoid the extreme points that may be used to distort the analysis, Interquartile Range (IQR) method was used to establish acceptable lower and upper limits. Any values that were beyond

these limits were substituted with the median of the respective feature. The median replacement was selected due to its resistance to extreme values and the ability to retain the central tendency of the data without excluding possibly useful records. After handling outliers, updated box plots shown in *Figure4* below showed a more reasonable and consistent data distribution across numerical features.

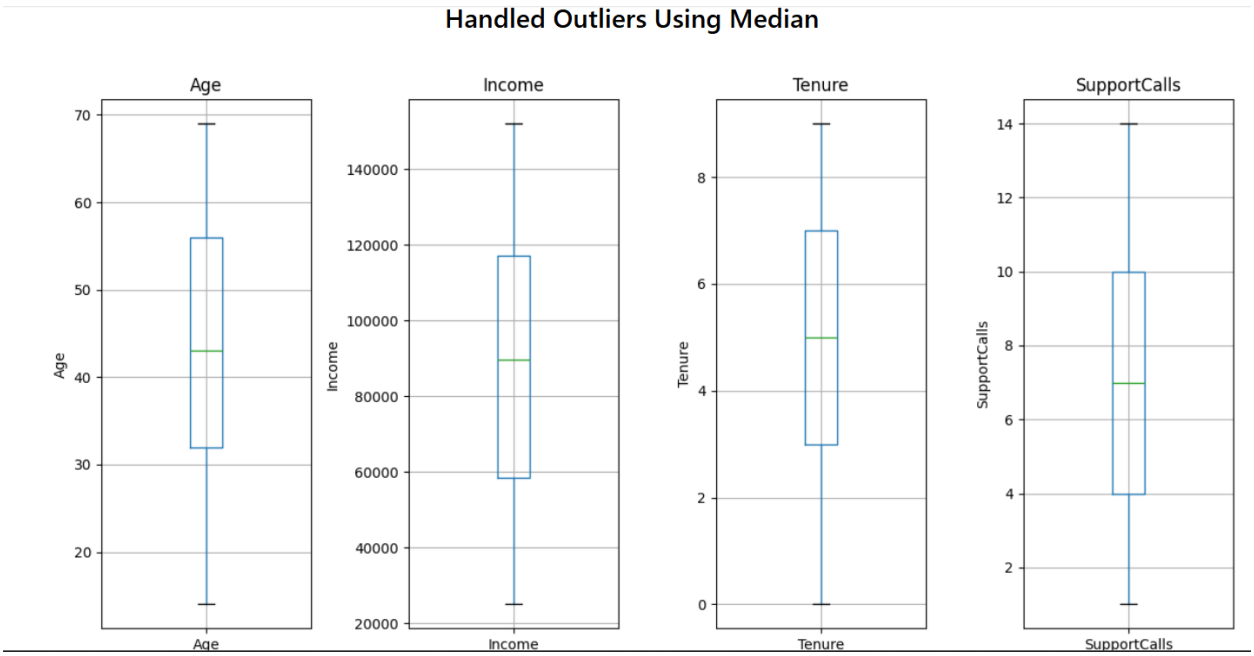


Figure 4: Handled Outliers using Median

Box plots before cleaning revealed clear extreme outliers, especially in Income and SupportCalls, which could negatively affect analysis and model performance. Outliers were treated using the IQR method by replacing values outside acceptable bounds with the median, helping preserve the central tendency of the data. After handling outliers, the box plots show no extreme points, resulting in more stable and realistic distributions that improve the dataset's reliability for further analysis.

4. Feature Scaling

To ensure that all numerical features contribute equally during modeling, Standardization (Z-score scaling) was applied to the dataset. This method transforms numerical variables such as Age, Income, Tenure, and SupportCalls to have a mean of 0 and a standard deviation of 1. Standardization is preferred since features vary greatly in scale for example, Income values are in the thousands while SupportCalls are small integers which could otherwise bias distance-based or gradient-based algorithms. After scaling, all numerical features are on a comparable scale as shown in *Figure5* below, improving model stability, performance, and training efficiency.

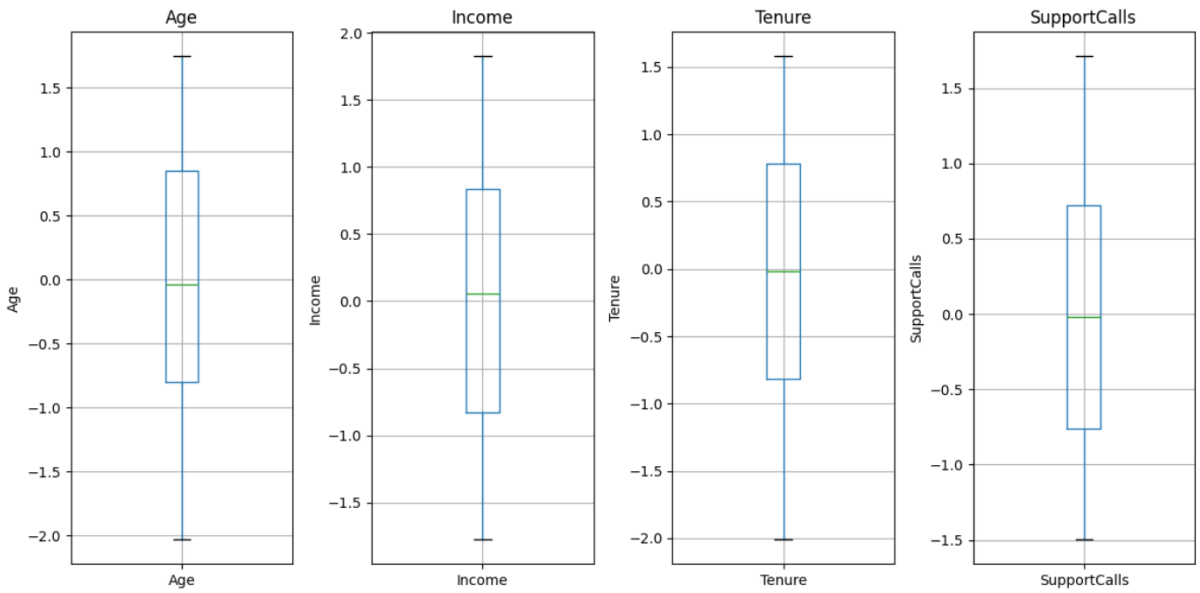


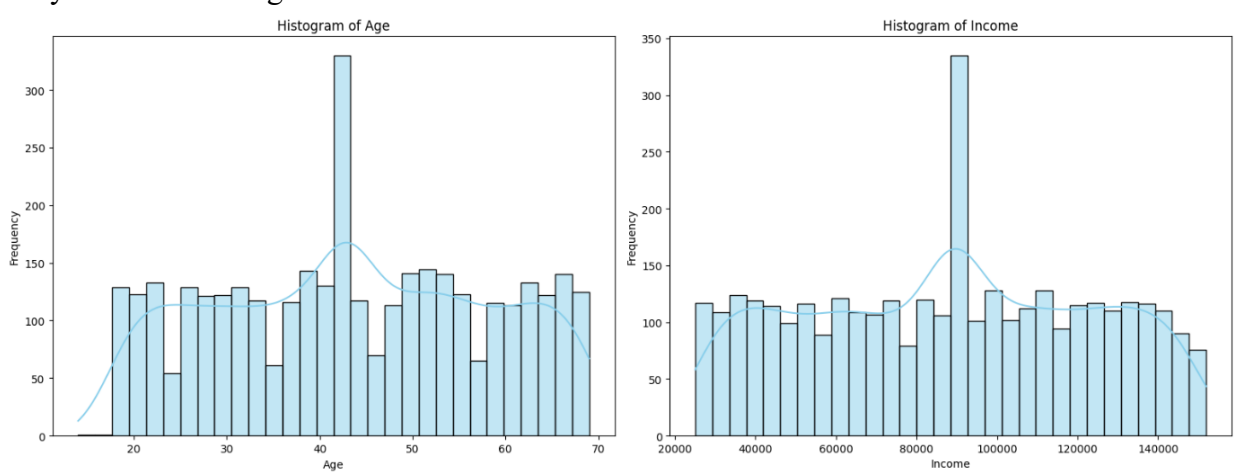
Figure 5: Feature Scaling using Z-score - Standardized Data

5. Exploratory Data Analysis (EDA):

5.1. Univariate Analysis:

5.1.1. Visualize the distribution of key numerical features using histograms

Histograms shown in *Figure6* below show the distributions of Age, Income, Tenure, and SupportCalls after standardization. All features are centered around 0, with most values within ±2 standard deviations. The plots reveal patterns in customer demographics and behavior, highlight relative variations, and help identify potential outliers, providing insights for further analysis and modeling.



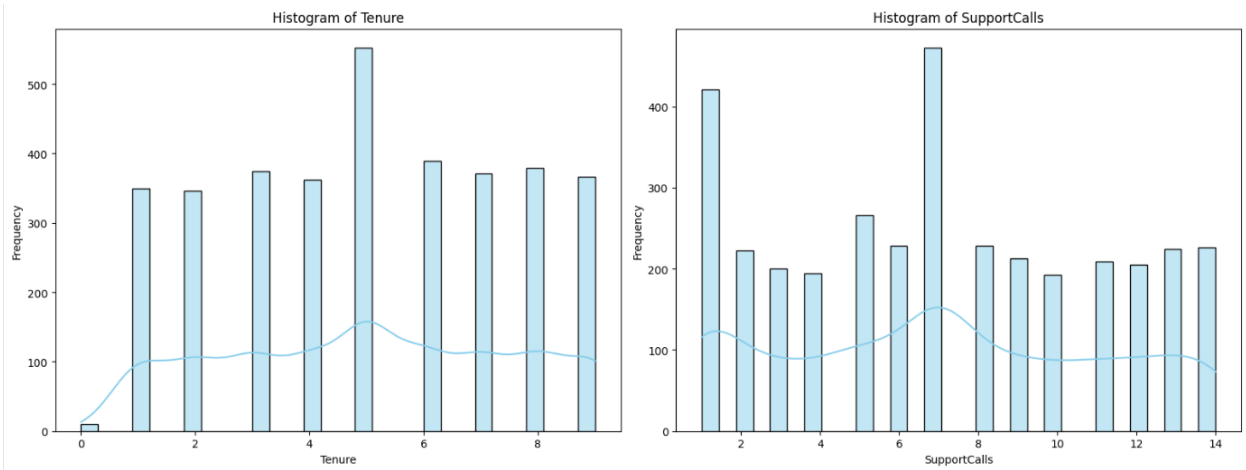


Figure 6: Univariate Analysis: Visualize the distribution of key numerical features using histograms

5.1.2. Analyzing the distribution of categorical variables using bar plots

Bar plots were used to examine the distribution of key categorical features: Gender, ProductType, and ChurnStatus. The plots shown in *Figure7* below show the proportion of male and female customers, the number of customers subscribed to Basic versus Premium products, and the count of customers who churned versus those who stayed. These visualizations highlight potential patterns in customer demographics and product preferences for example, any imbalance in churned versus stayed customers can inform modeling strategies and evaluation metrics.

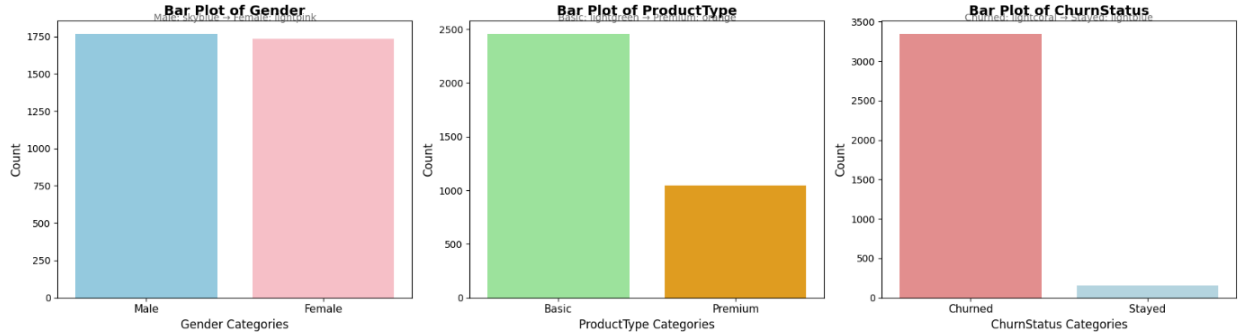


Figure 7: Analyzing the distribution of categorical variables using bar plots

5.1.2. Bivariate Analysis

5.1.2.1. Exploring relationships between numerical features and the target variable (ChurnStatus) using box plots.

Figure8 shown below shows box plots used to compare numerical features between customers who churned and those who stayed. The plots reveal patterns such as higher support calls and shorter tenure being associated with churn, while features like age and income show less pronounced differences. These insights help identify which numerical features may be most predictive of churn.

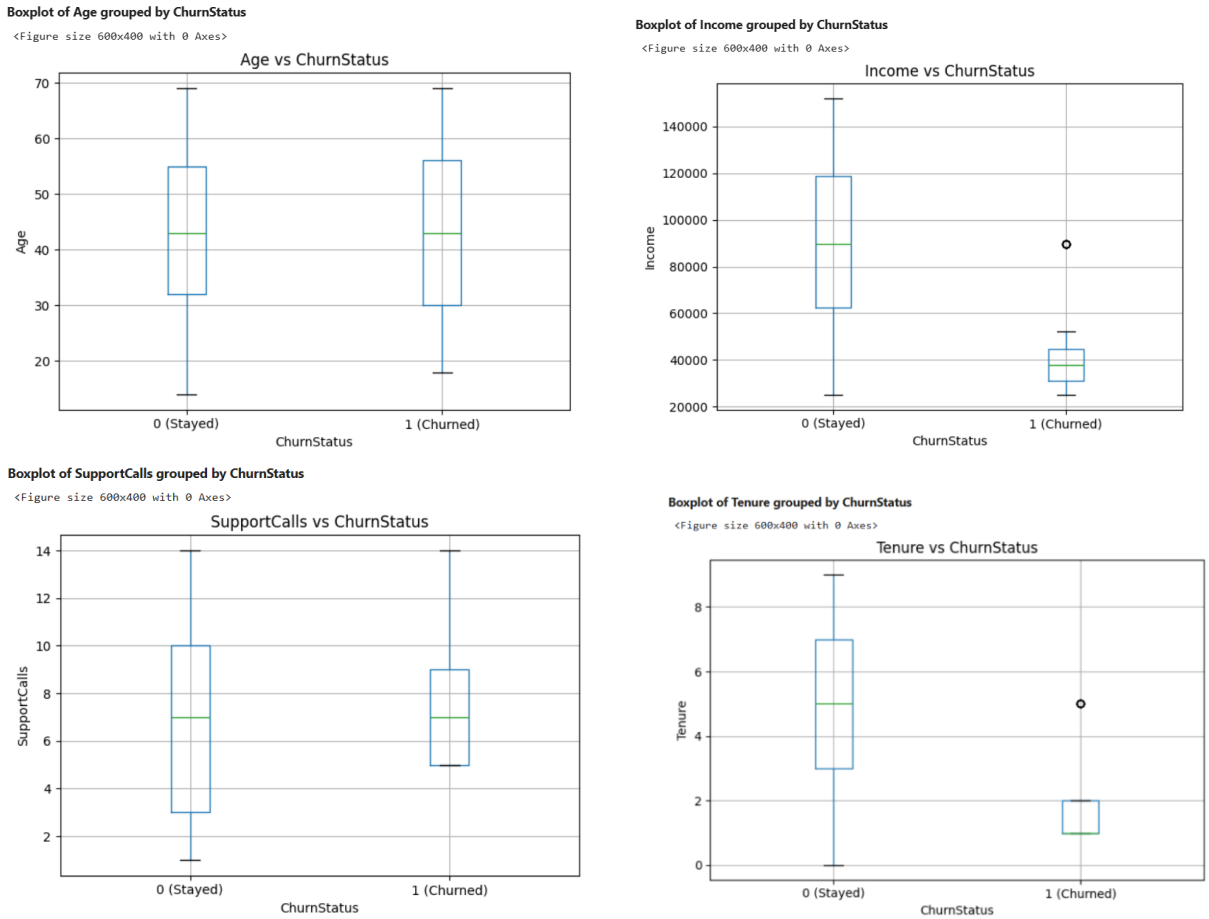


Figure 8: Bivariate Analysis: relationships between numerical features and the target variable (ChurnStatus) using box plots.

5.1.2.2. Investigating relationships between categorical variables and the target using bar plots

Stacked bar plots were used to explore the relationship between categorical features (Gender and ProductType) and ChurnStatus. Each bar represents the proportion of customers who stayed or churned within each category helping identify categorical factors that are more predictive of churn.

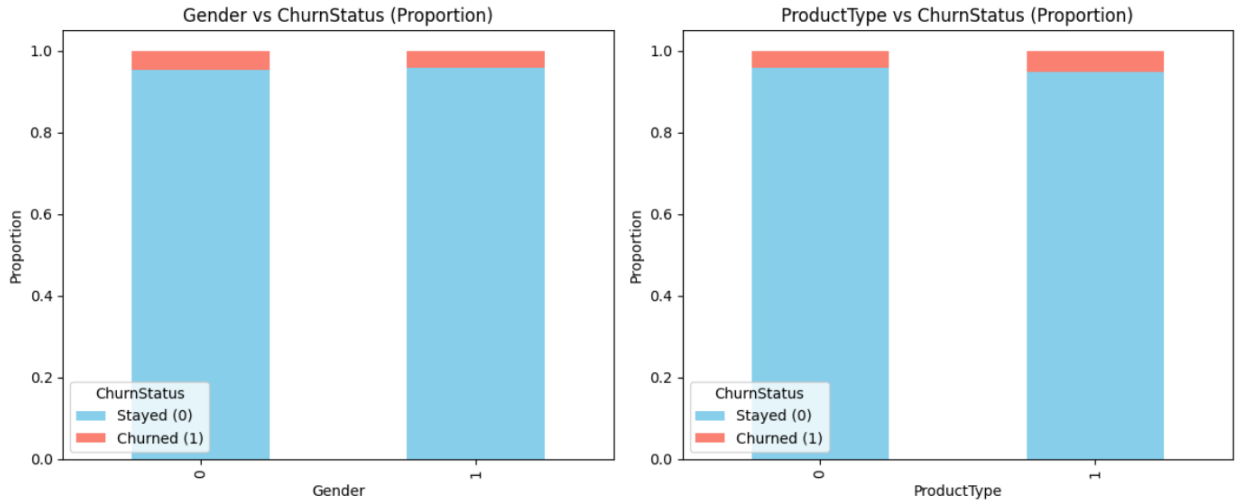


Figure 9: Bivariate relationships between categorical variables and the target using bar plots

5.3 Correlation Analysis

5.3.1. Creation of a correlation matrix to examine how numerical features relate t each other and to the target variable

A correlation matrix was created as shown in *Figure10* below to examine the relationships among numeric features (Age, Income, Tenure, SupportCalls) and the target variable (ChurnStatus). The heatmap shows the strength and direction of linear relationships between features, with values ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation). Features with higher absolute correlation values with ChurnStatus may be more predictive of churn, while correlations among independent features can help identify multicollinearity or redundancy.

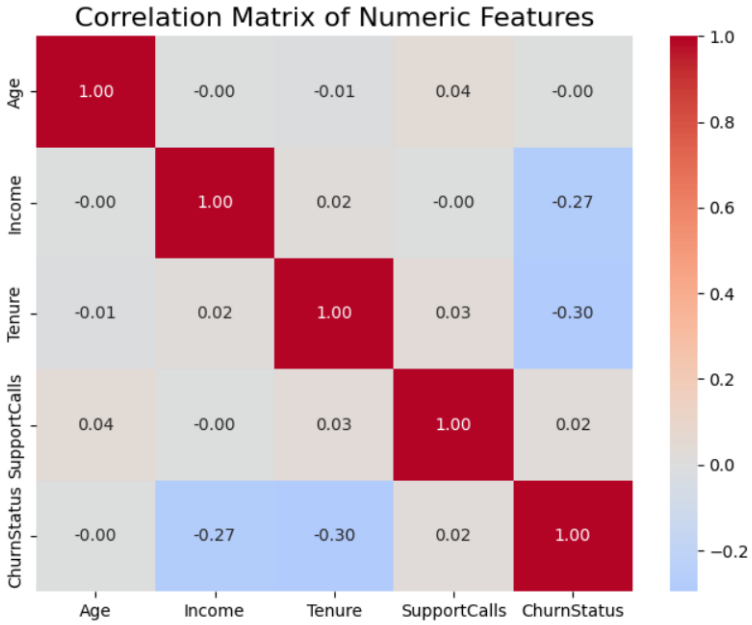


Figure 10: Correlation Analysis: Correlation Matrix

5.3.2. Discussing which features seem most predictive of churn

The *Figure10* shown above or results shown in *Figure11* below illustrate the correlation of numeric features with ChurnStatus. Both indicates which features are most predictive of customer churn. In this dataset, Tenure (-0.296) and Income (-0.271) show the strongest negative correlations with churn, suggesting that customers with higher income and longer tenure are less likely to churn. SupportCalls has a very weak positive correlation (0.024), while Age shows almost no correlation (-0.002) with churn. This analysis suggests that Tenure and Income are the most relevant numeric features for predicting customer churn in this dataset.

```
Correlation of features with ChurnStatus:

ChurnStatus     1.000000
SupportCalls    0.024181
Age            -0.001847
Income         -0.270773
Tenure         -0.295940
Name: ChurnStatus, dtype: float64
```

Figure 11: Correlation of features with ChurnStatus

6. Data Visualizations

In this part we visualize the 4 most critical relations between data, and where are the dependency and correlation actually live.

Visualization#1: Tenure vs Income by churn status:

It is represented using scatter plot, this figure shows the down square in the left down corner which insight us to the relation between tenure, income and churn status, the figure shows that when both tenure and income are low the churn status probability is high and very low otherwise.

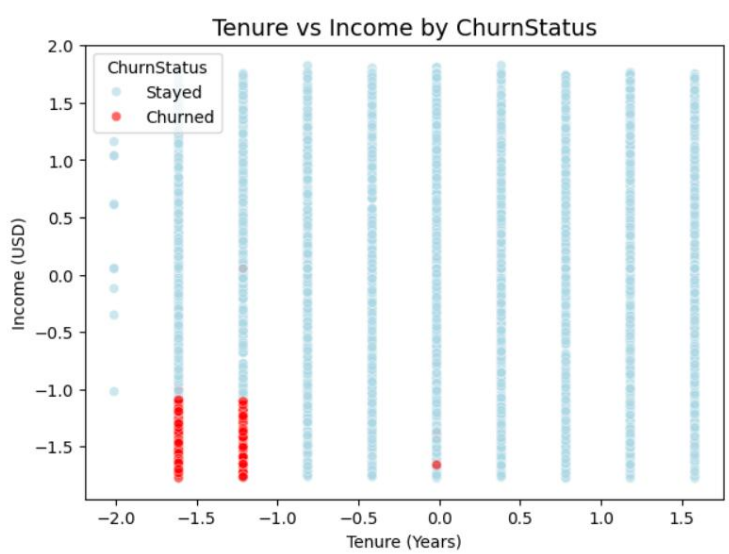


Figure 12: Data Visualization: Tenure vs Income by churn status

Visualization#2: ChurnStatus vs SupportCalls

It is represented using boxplot, since the highest positive correlation is between these two features, we insight that the more the customer make support calls the more likely he will churn.

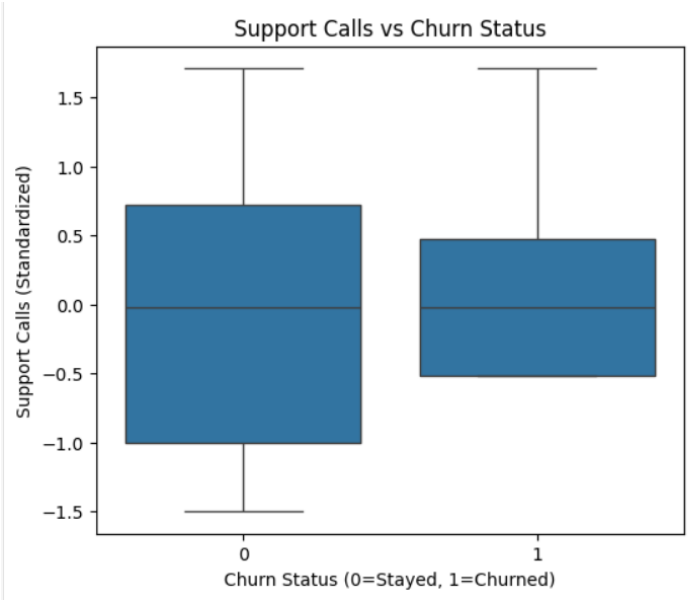


Figure 13: Data Visualization: ChurnStatus vs SupportCalls

Visualization#3: ChurnStatus vs Tenure

It is represented using boxplot, since the highest negative correlation is between these two features, we insight that the long-term customers with the highest tenure values are the least likely to churn.

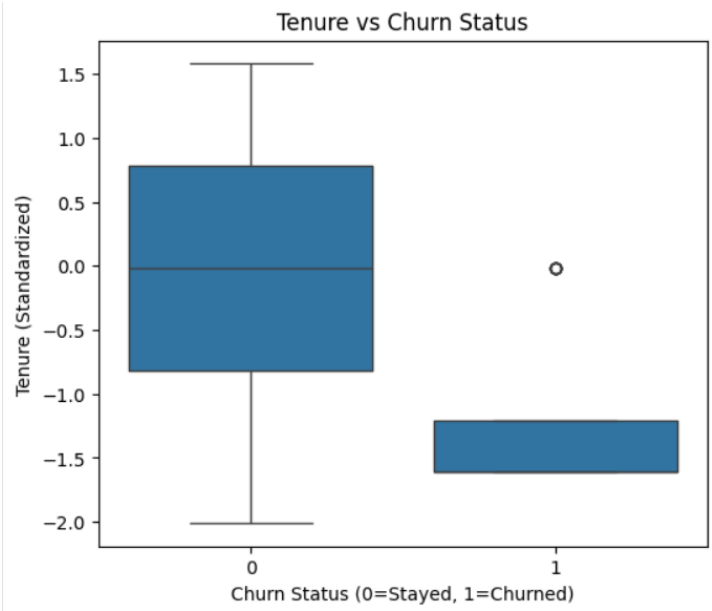


Figure 14: Data Visualization: ChurnStatus vs Tenure

Visualization#4: ChurnStatus vs Gender & ChurnStatus vs ProductType

we represented them using bar plots, from Figures15 shown below we insight that gender and product type (binary categories we have) don't affect the output as much as other features and their correlation is very close to 0, which means they don't have an effect on customers that churned since we can see that the number of male customers churned is almost as female customers and same for basic and premium product type.

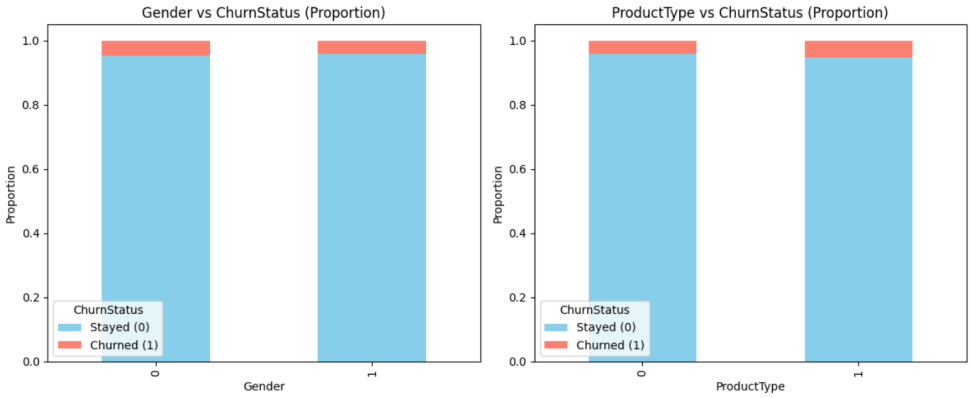


Figure 15: Data Visualization: ChurnStatus vs Gender & ChurnStatus vs ProductType

Key insights from EDA:

- Univariate analysis revealed the distributions of numeric and categorical features.
- Bivariate analysis highlighted differences in numeric features and proportions across churned vs. stayed customers.
- Correlation analysis showed that **Tenure** and **Income** are the most predictive numeric features for churn.
- Categorical features like **Gender** and **ProductType** showed no effect on churn proportions.

Significant Patterns and Insights:

- **Tenure vs Churn:** Customers with longer tenure are less likely to churn. Retaining long-term customers may require less effort, while new customers may need more engagement.
- **Income vs Churn:** Higher-income customers tend to stay, indicating that targeted offers or loyalty programs for lower-income customers could reduce churn.
- **SupportCalls vs Churn:** Slightly higher support calls may indicate potential dissatisfaction, but the correlation is weak. Monitoring frequent support requests could help identify at risk customers.
- **ProductType vs Churn:** Some product types may have higher churn rates; understanding product preferences can help tailor retention strategies.
- **Gender vs Churn:** Minor differences exist in churn proportions; while not highly predictive, gender-based insights could support marketing strategies.
- **Actionable Insights:** Focus retention strategies on newer, lower-income customers and monitor high-support-call customers to proactively reduce churn. These insights can also improve predictive models by emphasizing the most relevant features.

Conclusion

In this assignment we started with cleaning the data, fixing missing values, handling outliers, and then putting the numeric columns on the same scale using Z-score standardization. simple checks showed the scaling worked (means around 0 and standard deviations around 1). In Conclusion, tenure had the clearest link to churn (short-tenure customers churn more), while income showed a small negative link and support calls and age were weaker on their own. Another view suggested that new customers who contact support a lot are at higher risk than long-tenure customers with the same call volume. In the end, we visualized the data using several methods in order to conclude these insights from it.